

MANAGEMENT OF ASTHMA EXACERBATION IN THE CHILDREN'S EMERGENCY

ASSESS SEVERITY OF EXACERBATION

MILD – MODERATE

- SpO₂ >92%
- Ambulant
- Speaks in phrases and/or able to feed normally
- No features of severe episode

SEVERE

(any of the following)

- SpO₂ ≤92%
- Respiratory rate
 - <2 years >50 /min
 - 2–5 years >40 /min
 - >5 years >30 /min
- Heart rate
 - <5 years >140 /min
 - >5 years >125 /min
- Use of accessory neck muscles

LIFE THREATENING

- SpO₂ ≤92%
- (plus any of the following)
 - Altered consciousness
 - Silent chest
 - Cyanosis
 - Impending exhaustion

TREATMENT

- Oxygen to achieve SpO₂ >94%
- MDI Salbutamol via spacer 4–6 puffs every 20 minutes for 3 doses

TREATMENT

- Oxygen via nasal prongs or facemask to achieve SpO₂ >94%
- Nebulised Salbutamol plus Ipratropium
 - <5 years: 0.5ml 0.5% Salbutamol plus 2.5ml 0.9% NaCl plus 1ml (250mcg) Ipratropium every 20 minutes for 3 doses
 - ≥5 years: 1ml 0.5% Salbutamol plus 2ml 0.9% NaCl plus 1ml (250mcg) Ipratropium every 20 minutes for 3 doses
- Oral Prednisolone 1mg/kg/day (max 40mg/dose) once daily or IV Hydrocortisone 4mg/kg/dose (max 200mg/dose) 6 hourly if child is not feeding well

TREATMENT

- Inform duty Intensivist and arrange transfer to PICU
- High flow oxygen to achieve SpO₂ >94%
- Continuous Nebulised Salbutamol 1ml 0.5% Salbutamol diluted with 2ml 0.9% NaCl hourly (Consider neat salbutamol if suboptimal response) Add Neb Ipratropium 1ml (250mcg) every 20 minutes for 3 doses, then 4–6 hourly
- IV Hydrocortisone 4mg/kg/dose (max 200mg/dose) 6 hourly
- Consider:
 - IV Magnesium sulphate 50% 0.1ml/kg (max 4ml/dose) in 10ml 0.9% NaCl over 20 minutes 6 hourly
 - Intubation and Mechanical ventilation
 - Chest x-ray and/or Blood gases

REASSESS within 1 hour

INCOMPLETE RESPONSE

- MDI Salbutamol via spacer 4–6 puffs every 20 minutes for 3 doses
- Consider:
 - PO Prednisolone 1mg/kg/day (max 40mg/dose) once daily

REASSESS within 1 hour

GOOD RESPONSE

- Resolution of wheeze
- No respiratory distress
- SpO₂ >94% in room air

INCOMPLETE RESPONSE

- Tachypnoea / wheeze not improving
- Signs of respiratory distress
- SpO₂ ≤94% in room air

POOR RESPONSE

- Worsening respiratory status
- Continuing use of accessory neck muscles
- Any life threatening features

DISCHARGE PLAN

- Continue Salbutamol via MDI and spacer 4–6 puffs 4 hourly for 3 days and as needed thereafter
- Ensure parents are confident in caring for child
- Review asthma action plan
- Assess inhaler technique
- Follow-up with GP or asthma clinic within 1 week to consider need for inhaled steroids
- Inform asthma nurse in the following:
 - Previous ICU admission
 - Poor compliance with inhaled steroids
 - Re-attendance to CE in the last 3 months
 - Social concerns

ADMIT TO GENERAL WARD

- Avoid admission to isolation ward (Ward 62), unless strongly suspecting contagious disease
- Refer to inpatient management guidelines

References:

- 1) GINA report, 2009
- 2) BTS/SIGN guidelines on the management of asthma, 2010
- 3) NHLBI expert panel report, 2009
- 4) PRACTALL consensus report, 2008

Disclaimer: This document reflects what is currently regarded as safe practice. However, as in any clinical situation, there may be factors which cannot be covered by a single set of guidelines. This document does not replace the need for the application of clinical judgement to each individual presentation.